

9. Conditional Statements



-  **if else**
✓ video 10:58
-  **Print the largest of 2 numbers**
✓ video 04:56
-  **Print if number is Odd or Even**
✓ video 04:23
-  **else if**
✓ video 03:00
-  **Income Tax Calculator**
✓ video 06:56
-  **Print the largest of 3 numbers**
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-  **Ternary Operator**
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✓ video 01:55
-  **Switch statement**
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-  **Calculator**
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-  **Practice Questions**
✓ pdf

Conditional Statements



6
↓
alarm ring ✓



daant ✓
↓
saaf ✓



> 33 ✓
↓
pass ✓

< 33
↓
fail

condⁿ - sach
↓
kaam



Conditional Statements

if, else

else if

✓ ternary operator

switch



if-else Statements

```
if (condition) {
```

```
}
```

```
else {
```

```
}
```



JavaBasics.java > JavaBasics > main(String[])

2

3 public class JavaBasics {

Run | Debug

4 public static void main(String args[]) {

5 int age = 16;

6 if(age >= 18) {

7 System.out.println("adult : drive, vote");

8 }

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10

11 else {

12 System.out.println("not adult");

13 }

14

15 }

16 }

17



Print the largest of 2

A = 1, B = 3

A >= B ~~if~~ true → print "A"

~~else~~ false → print "B"

```
if ( A >= B ) {  
    syso("A");  
}  
else {  
    syso("B");  
}
```



JavaBasics.java 1 X

JavaBasics.java > JavaBasics > main(String[])

Run | Debug

```
4 public static void main(String args[]) {  
5     int A = 10;  
6     int B = 5;  
7  
8     if(A >= B) {  
9         System.out.println("A is largest of 2");  
10    } else {  
11        System.out.println("B is largest of 2");  
12    }  
13  
14 }
```

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PROBLEMS

1

OUTPUT

TERMINAL

DEBUG CONSOLE

```
shradhakhapra@Shradhas-Air Java Programming Language % java JavaBasics  
B is largest of 2  
shradhakhapra@Shradhas-Air Java Programming Language % java JavaBasics.  
A is largest of 2  
shradhakhapra@Shradhas-Air Java Programming Language %
```



a b

 $8/2$ $4/2$ $10/2$

Print if a number is Odd or Even

4 (even)

divisible by 2 ✓
or
multiple of 2 ✓

1 (odd)



Print if a number is Odd or Even

4 (even)

```
if (n % 2 == 0)
    ↓
    print("EVEN")
else
    ↪ print("ODD")
```

$$a \% b = 0$$
$$\text{even} \% 2 = 0$$



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JavaBasics.java > JavaBasics > main(String[])

```
1  import java.util.*;
2
3  public class JavaBasics {
    Run | Debug
4      public static void main(String args[]) {
5          Scanner sc = new Scanner(System.in);
6          int number = sc.nextInt();
7
8          if(number % 2 == 0) {
9              System.out.println("EVEN");
10         } else {
11             System.out.println("ODD");
```

PROBLEMS 1 OUTPUT TERMINAL DEBUG CONSOLE

shradhakhapra@Shradhas-Air Java Programming Language % java JavaBasics

2

EVEN

shradhakhapra@Shradhas-Air Java Programming Language %



else if Statements

```
if (condition 1) {
```

```
}
```

```
else if (condition 2) {
```

```
}
```

```
else {
```

```
}
```

```
if (cond1) {
```

```
}
```

```
if (cond2) {
```

```
}
```

```
...
```



Income Tax Calculator

income < 5L

0% tax

income between 5-10L

20% tax

income > 10L

30% tax



Income Tax Calculator

① income < 5L
0% tax ✓

income between 5-10L

```
int income (user)
int tax (calculate)
```

```
if (income < 5L) {
    tax = 0;
}
else if (income >= 5L && income < 10L)
    tax = income * (0.2)
}
else {
    tax = income * (0.3)
```



JavaBasics.java > JavaBasics > main(String[])

```
1  import java.util.*;
2
3  public class JavaBasics {
    Run | Debug
4  public static void main(String args[]) {
5      Scanner sc = new Scanner(System.in);
6      int income = sc.nextInt();
7      int tax;
8
9      if(income < 500000) {
10         tax = 0;
11     }
12     else if(income >= 500000 && income < 1000000) {
13         tax = (int) (income * 0.2);
14     }
15     else {
16         tax = (int) (income * 0.3);
17     }
18
19     System.out.println("your tax is : " + tax);
20
```



Print the largest of 3

A = 1, B = 3, C = 6

```
if (A >= B) && (A >= C) {  
    print("A")
```

```
}
```

```
else if (B >= C) {  
    print("B")
```

```
}
```

```
else print("C")
```



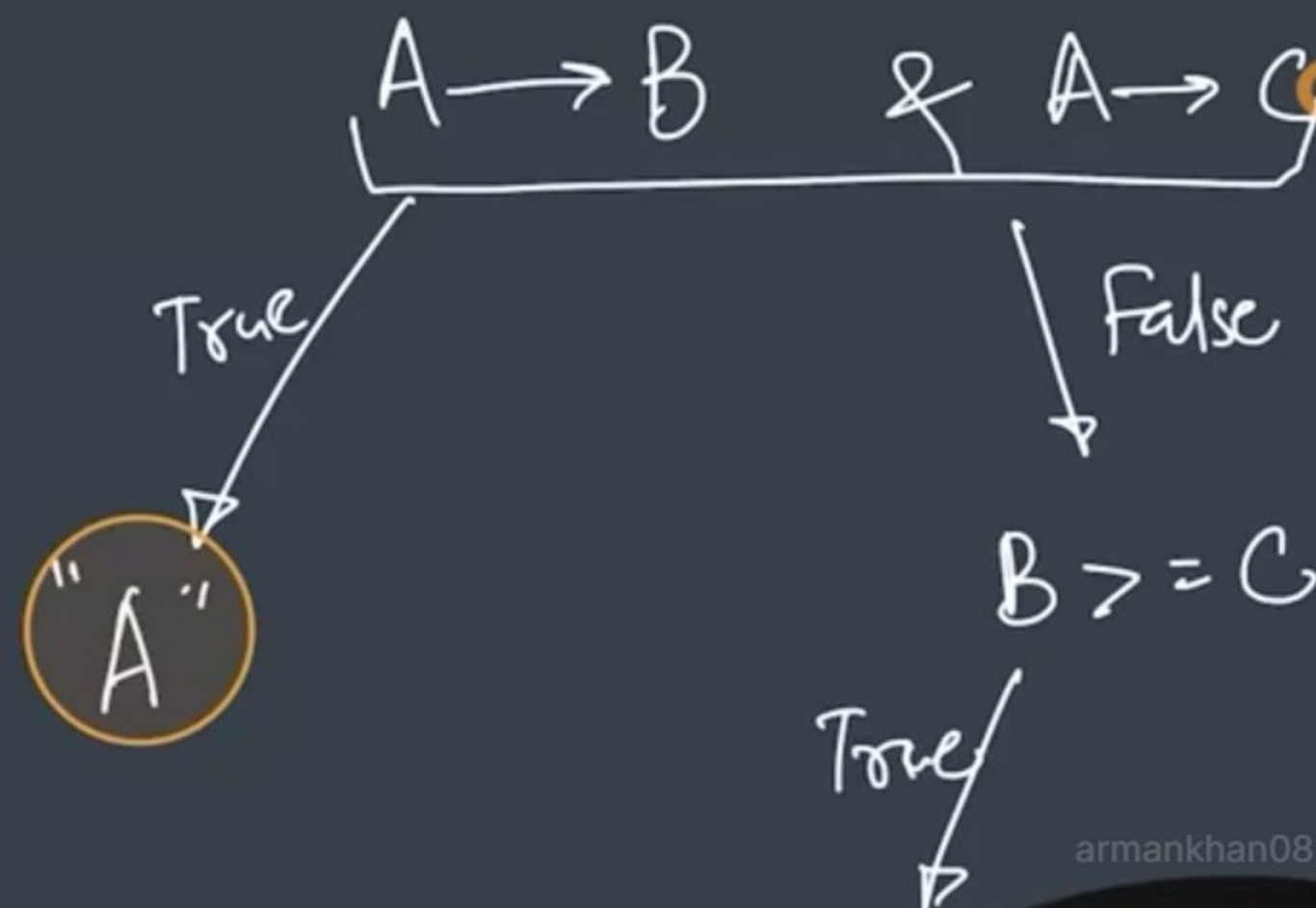
Print the largest of 3

A = 1, B = 3, C = 6

```
if (A >= B) && (A >= C) {
    print("A")
}
```

```
else if (B >= C) {
    print("B")
}
```

```
else {
    print("C")
}
```



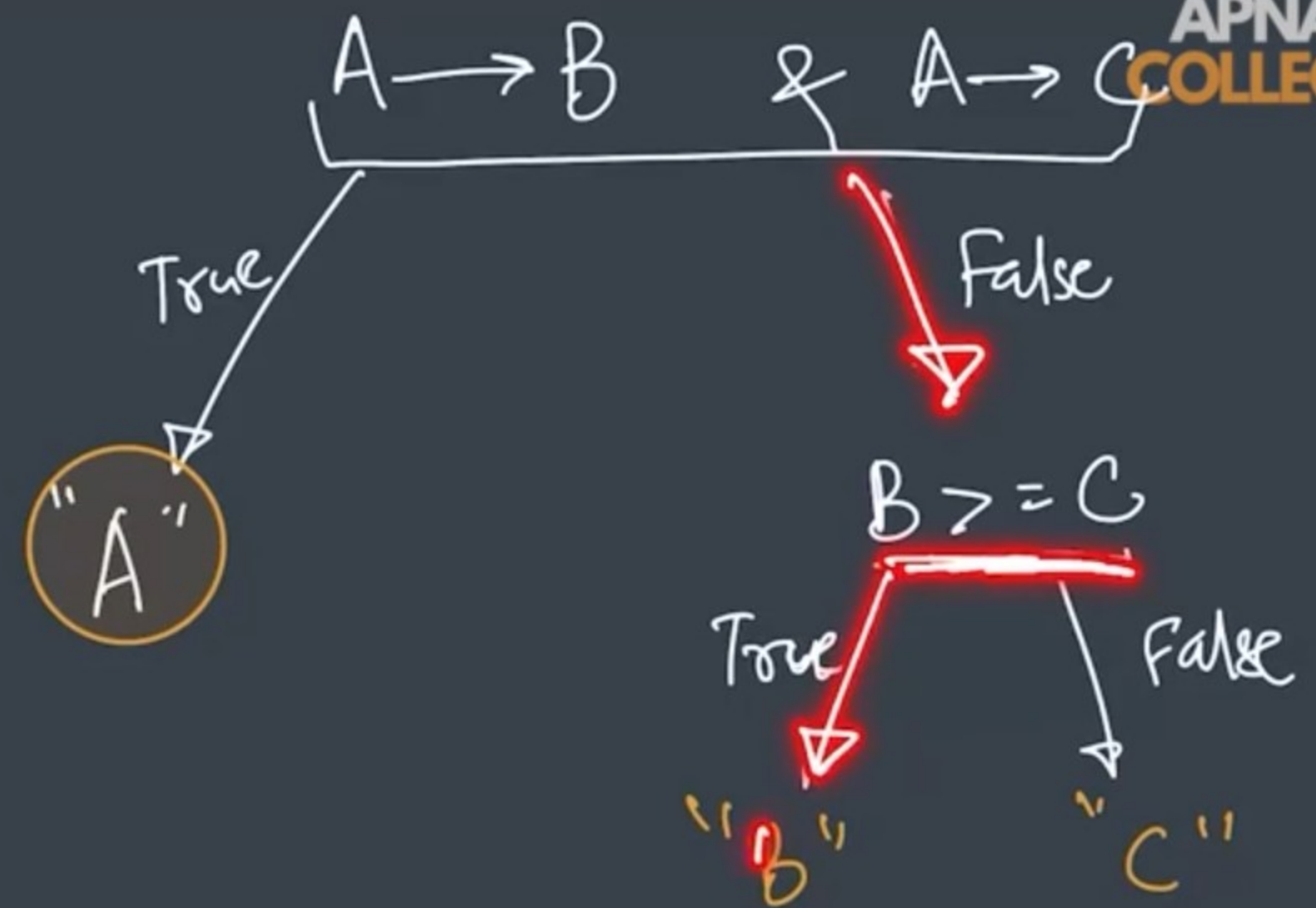
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Print the **largest of 3**

A = 1, B = 3, C = 6

```
if (A >= B && (A >= C)) {  
    print("A")  
}
```



JavaBasics.java > JavaBasics > main(String[])

```
1  import java.util.*;
2
3  public class JavaBasics {
    Run | Debug
4      public static void main(String args[]) {
5          int A = 1, B = 3, C = 6;
6
7          if( (A >= B) && (A >= C)) {
8              System.out.println("largest is A");
9          }
10         else if(B >= C) {
11             System.out.println("largest is B");
12         }
13         else {
14             System.out.println("largest is C");
15         }
16     }
17 }
18
```



ternary Operator

variable $\equiv$ condition? statement1 : statement2;

Handwritten notes:
- Above the colon: नही तो (Not then)
- Below 'true': true
- Below 'false': false
- Red arrows: One from 'condition?' to 'statement1', and another from 'statement1' to 'statement2'.

boolean larger = (5 > 3)? 5 : 3;

String type = (5 % 2 == 0)? "even" : "odd";



JavaBasics.java > JavaBasics > main(String[])

```
3 public class JavaBasics {
```

Run | Debug

```
4     public static void main(String args[]) {
```

```
5         int number = 5;
```

```
6
```

```
7         //ternary operator
```

```
8         String type = ((number%2) == 0) ? "even" : "odd";
```

```
9         System.out.println(type);
```

```
10    }
```

```
11 }
```

```
12
```

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PROBLEMS

1

OUTPUT

TERMINAL

DEBUG CONSOLE

```
shradhakhapra@Shradhas-Air Java Programming Language % java JavaBasics.  
even
```

```
shradhakhapra@Shradhas-Air Java Programming Language % java JavaBasics.  
odd
```

```
shradhakhapra@Shradhas-Air Java Programming Language %
```



```
3 public class JavaBasics {  
    Run | Debug  
4     public static void main(String args[]) {  
5         int marks = 67;  
6  
7         String reportCard = marks >= 33 ? "PASS" : "FAIL";  
8         System.out.println(reportCard);  
9     }  
10 }  
11
```

I



Switch Statement



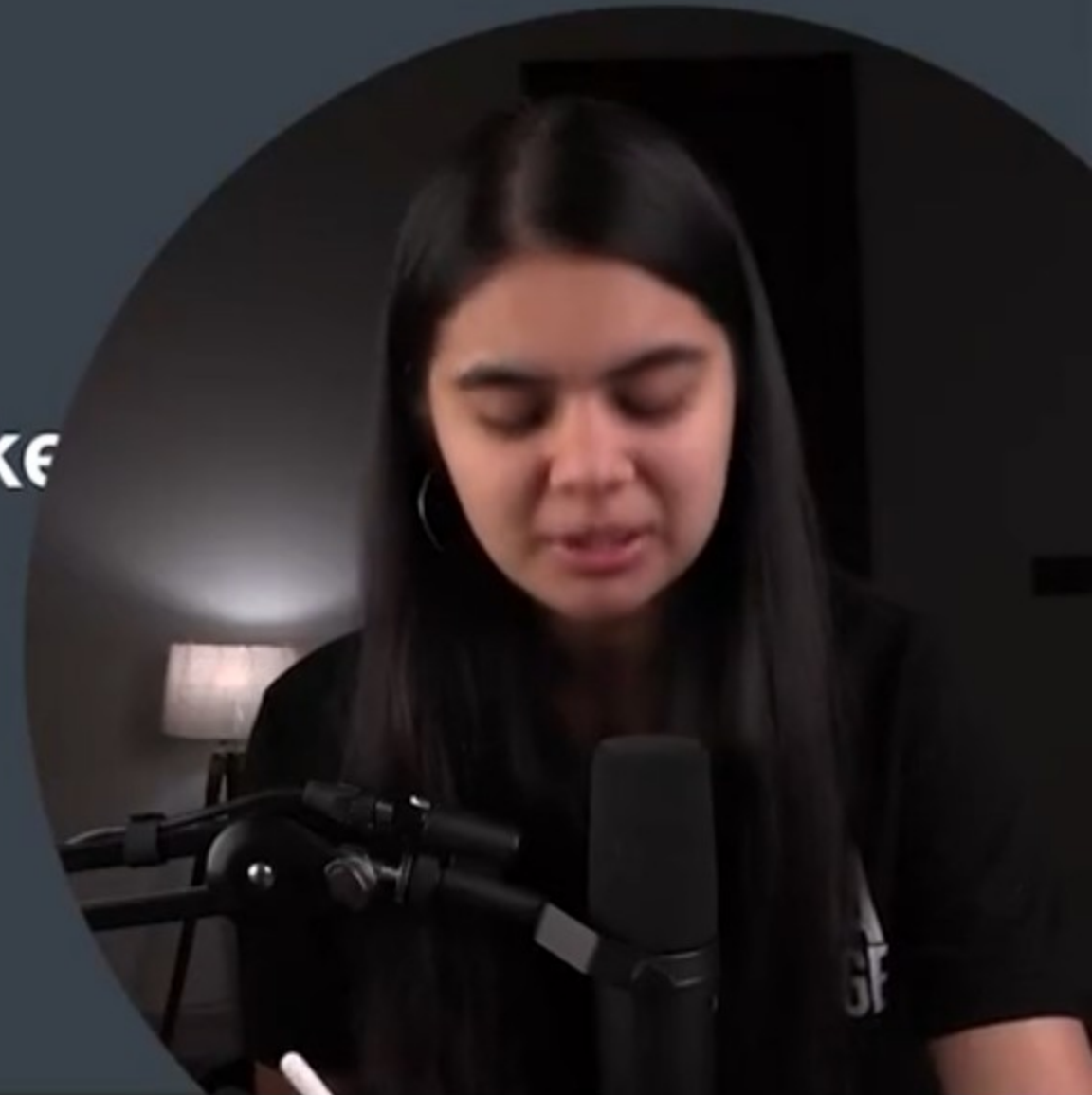
Samosa



Burger



Mango Shake





Switch Statement

switch (variable) {

case 1 :

case 2 :

case 3 :

default :

}



JavaBasics.java > JavaBasics > main(String[])

```
4 public static void main(String args[]) {  
5     int number = 2;  
6     switch(number) {  
7         case 1 : System.out.println("samosa");  
8             break;  
9         case 2 : System.out.println("burger");  
10            break;  
11        case 3 : System.out.println("mango shake");  
12            break;  
13        default : System.out.println("we wake up");  
14    }  
15 }
```

PROBLEMS 1 OUTPUT TERMINAL DEBUG CONSOLE

```
shradhakhapra@Shradhas-Air Java Programming Language % java JavaBasics.  
burger  
mango shake  
we wake up  
shradhakhapra@Shradhas-Air Java Programming Language %
```

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```
3 public class JavaBasics {  
    Run | Debug  
4     public static void main(String args[]) {  
5         char ch = 'a';  
6         switch(ch) {  
7             case 'd' : System.out.println("samosa");  
8                 break;  
9             case 'a' : System.out.println("burger");  
10                break;  
11             case 'c' : System.out.println("mango shake");  
12                break;  
13             default : System.out.println("we wake up");  
14         }  
15     }  
16 }  
17
```



```
3 public class JavaBasics {  
    Run | Debug  
4     public static void main(String args[]) {  
5         Scanner sc = new Scanner(System.in);  
6         int a = sc.nextInt();  
7         int b = sc.nextInt();  
8         char operator = sc.next().charAt(0);  
9  
10        switch(operator) {  
11            case '+': System.out.println(a+b);  
12                    break;  
13            case '-': System.out.println(a-b);  
14                    break;  
15            case '*': System.out.println(a*b);  
16                    break;  
17            case '/': System.out.println(a/b);  
18                    break;  
19            case '%': System.out.println(a%b);  
20                    break;  
21            default : System.out.println("wrong operator");  
22        }  
23    }
```

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